

CYBERSECURITY INTERNSHIP

Enumeration — Digging Deeper into Services

Enumeration Report — Week 3, Task 9

Target Host: 192.168.139.129

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Task	Week 3, Task 9: Enumeration — Digging Deeper into Services
Attacker Machine	TryHackMe
Date Submitted	September 7, 2026

1. Objective

This report documents active service enumeration performed against a target host, focused on extracting usernames, shares, and service versions from SMB, FTP, and SSH. The goal was to move beyond simple port discovery covered in earlier tasks and pull actionable, human-readable information from each open service using enum4linux, smbclient, and manual banner grabbing with netcat.

Target host selected: 192.168.139.129 (Metasploitable 2).

The attacker machine (Kali Linux) and target sit on the same host-only virtual subnet established in earlier Nmap discovery tasks of this internship track.

2. SMB Enumeration — smbclient

Command used:

```
smbclient -L 192.168.139.129 -N
```

The -N flag suppresses the password prompt for a null/anonymous session. The target accepted the anonymous login without credentials, which is itself a security finding.

```
(mahnoor@kali)-[~/Desktop/techbiz]
$ smbclient -L 192.168.139.129 -N
Anonymous login successful

Sharename      Type      Comment
-----
print$         Disk      Printer Drivers
tmp            Disk      oh noes!
opt            Disk
IPC$           IPC       IPC Service (metasploitable server (Samba 3.0.20-Debian))
ADMIN$         IPC       IPC Service (metasploitable server (Samba 3.0.20-Debian))

Reconnecting with SMB1 for workgroup listing.
Anonymous login successful

Server          Comment
-----
Workgroup       Master
WORKGROUP      METASPLOITABLE

(mahnoor@kali)-[~/Desktop/techbiz]
$ nc 192.168.139.129 21
(UNKNOWN) [192.168.139.129] 21 (ftp) : Connection refused
```

Figure 1: smbclient anonymous share listing on 192.168.139.129

Item	Value
Samba version	3.0.20-Debian
Workgroup	WORKGROUP
Master browser	METASPLOITABLE
Shares found	print\$, tmp, opt, IPC\$, ADMIN\$
Auth	Anonymous / null session accepted

2.1 SMB Username Enumeration & Session Check — enum4linux

Command used:

```
enum4linux -a 192.168.139.129
```

The -a flag runs the full check set: workgroup/domain identification, NetBIOS status, null session testing, domain SID lookup, and RID-cycling based username enumeration.

```

(mahnoor@kali)-[~/Desktop/techbiz]
$ enum4linux -a 192.168.139.129
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Mon Sep  7 14:37:14 2026

===== ( Target Information ) =====
Target ..... 192.168.139.129
RID Range ..... 500-550,1000-1050
Username ..... ''
Password ..... ''
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none

===== ( Enumerating Workgroup/Domain on 192.168.139.129 ) =====
[+] Got domain/workgroup name: WORKGROUP

===== ( Nbtstat Information for 192.168.139.129 ) =====
Looking up status of 192.168.139.129
METASPLOITABLE <00> - B <ACTIVE> Workstation Service
METASPLOITABLE <03> - B <ACTIVE> Messenger Service
METASPLOITABLE <20> - B <ACTIVE> File Server Service
.._MSBROWSE_ <01> - <GROUP> B <ACTIVE> Master Browser
WORKGROUP <00> - <GROUP> B <ACTIVE> Domain/Workgroup Name
WORKGROUP <1d> - B <ACTIVE> Master Browser
WORKGROUP <1e> - <GROUP> B <ACTIVE> Browser Service Elections
MAC Address = 00-00-00-00-00-00

===== ( Session Check on 192.168.139.129 ) =====
[+] Server 192.168.139.129 allows sessions using username '', password ''

===== ( Getting domain SID for 192.168.139.129 ) =====
Domain Name: WORKGROUP
Domain Sid: (NULL SID)
[+] Can't determine if host is part of domain or part of a workgroup

```

Figure 2: enum4linux full enumeration output

Item	Value
Domain/Workgroup	WORKGROUP
Null session	Allowed (username "", password "")
Usernames enumerated	administrator, guest, krbtgt, domain admins, root, bin, none
Domain SID	(NULL SID) — standalone host, not domain-joined
NetBIOS services	Workstation, Messenger, File Server, Master Browser (all active)

3. FTP Banner Grab

Command used:

```
nc 192.168.139.129 21
```

Result: Connection refused. This is expected rather than an error — the xinetd process (which supervises vsftpd on this target) was intentionally killed during a prior hardening exercise, closing port 21 along with telnet (port 23). Prior to hardening, this port served vsftpd 2.3.4, notable for its backdoor vulnerability (CVE-2011-2523) identified during earlier vulnerability assessment work on this target.

4. SSH Banner Grab

Command used:

```
nc 192.168.139.129 22
```

```

(mahnoor@kali)-[~/Desktop/techbiz]
$ nc 192.168.139.129 22
SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1

```

Figure 3: Raw SSH banner returned on connect

Banner returned: SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1. OpenSSH 4.7p1 dates to 2007 and is well past end-of-life, though no unauthenticated RCE as severe as the FTP backdoor is known for this specific version.

5. TryHackMe Room Completions (Proof)

Both mandatory free-tier rooms listed in the task were completed:

Vulnversity

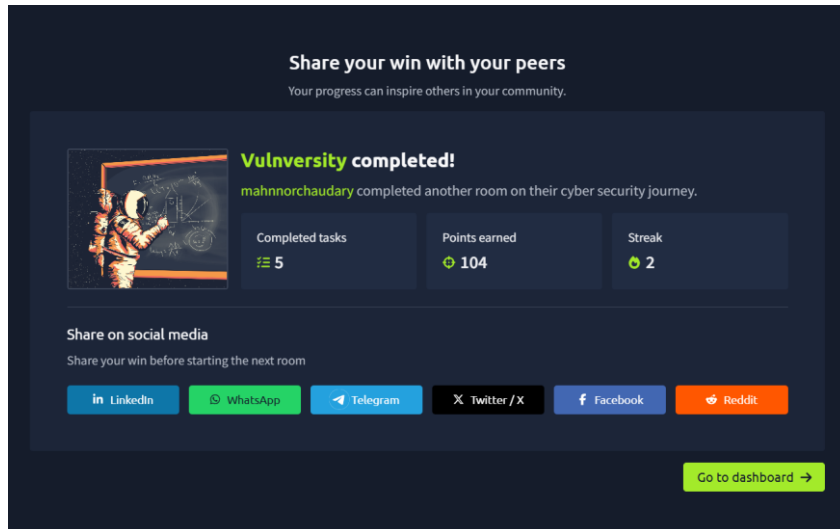


Figure 4: Vulnversity room completed — 5 tasks, 104 points

Nmap Room

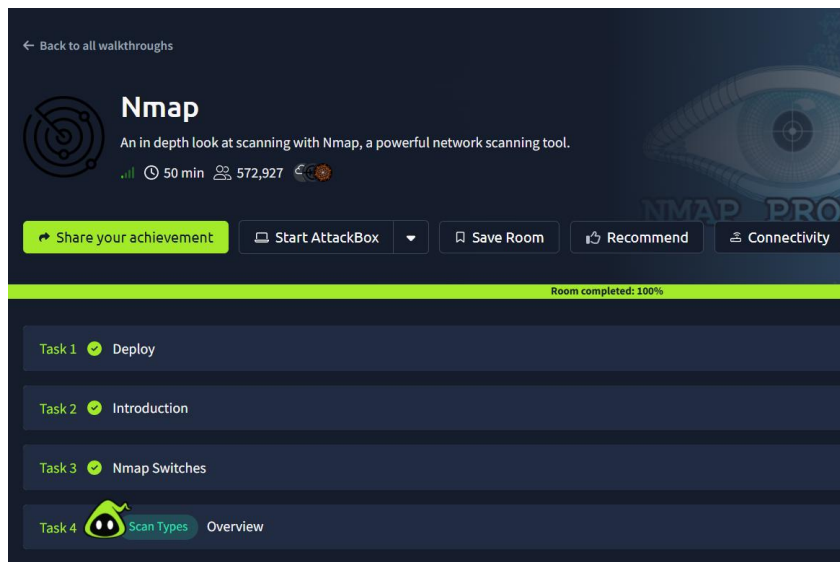


Figure 5: Nmap room completed — 100%

6. Summary of Findings

Service	Port	Version / Banner	Key Finding
SMB	139/445	Samba 3.0.20-Debian	Anonymous login, 5 shares, 7 usernames enumerated

Service	Port	Version / Banner	Key Finding
FTP	21	vsftpd 2.3.4 (pre-hardening)	Closed post-hardening; historically backdoored (CVE-2011-2523)
SSH	22	OpenSSH 4.7p1 Debian-8ubuntu1	Legacy, unsupported version; banner confirms host OS

7. Conclusion

Enumeration confirmed that SMB is the most information-rich service on this host: an unauthenticated anonymous session exposed the Samba version, all configured shares, and a usable username list via RID cycling — all without any credentials. FTP could not be banner-grabbed as it was deliberately disabled in a prior hardening step, and its historical vsftpd 2.3.4 backdoor remains documented from earlier vulnerability scanning. SSH exposed an outdated OpenSSH build via its banner. Together these results reinforce that null-session SMB access and legacy service versions remain the most practical entry points for further exploitation on this target.